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Separating Ownership and Information†

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This paper identifies an upside of the separation of ownership and control, typically the source of inefficiencies in the theory of the firm. Because insiders obtain private information by exercising control, the separation of ownership and control leads to a separation of ownership and information. We show that this separation is necessary for efficient trade in the market for corporate control. The analysis reveals how strategic communication between inside and outside shareholders facilitates takeovers by eliciting external bidders' private information. Our results call into question mandatory disclosure requirements during takeovers. (*JEL* D21, D82, G32, G34)

The separation of ownership and control is widely regarded as the fundamental source of inefficiencies in the firm because it creates agency costs (Berle and Means 1932; Jensen and Meckling 1976). This paper identifies an upside of such a separation: it can improve the allocation of control rights. Because insiders become privately informed by virtue of exercising control, the separation of ownership and control naturally leads to a *separation of ownership and information*. We show that this separation is necessary for efficient trade in the market for corporate control.

We present a model of corporate takeovers with two-sided, asymmetric information. An insider has private information regarding the target firm's stand-alone value, and a potential external bidder has private information regarding the target's value under his leadership. One can think of the insider as an actively involved shareholder or as the incumbent management. The majority of the target's shares are owned by uninformed, outside shareholders. The bidder can obtain control by acquiring a controlling stake of the target's shares through a tender offer. The insider can respond by a cheap talk recommendation to the outside shareholders. Finally, the outside shareholders observe the tender offer and the cheap talk recommendation, and then decide whether to tender their shares.

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The two-sided, asymmetric information implies that no entity by herself knows who should optimally obtain control. The bidder's tender offer gives rise to a signaling game in which both the outside shareholders' tendering decisions and the insider's communication strategy are endogenous to the tender offer. Moreover, the level of disagreement between the insider and outside shareholders and, therefore, the scope for communication are endogenous to the bidder's tender offer.

If ownership and control are not separated—that is, if the insider is the sole owner—she needs to sell part of her shares to induce a successful takeover. We show that, on the traded shares, the insider needs to share the post-takeover security benefits[1] with the bidder. By contrast, when keeping control, the insider enjoys all the security benefits she creates. This asymmetry endogenously biases her against the takeover such that not all value-increasing takeovers can be realized.

If ownership and control are separated, the efficient allocation of control rights can be achieved in equilibrium. Because the majority of shares are held by outside shareholders, the informed insider can retain her minority stake while relinquishing control to the bidder. In the efficient equilibrium, the insider does not tender any shares but limits herself to advising the outside shareholders on their tendering decisions. How can the insider's advice facilitate takeovers?

We identify a dual role of the insider's strategic recommendation. First, it provides part of the insider's private information. Second, it can be used to incentivize the bidder to fully reveal his private information via the tender offer. The insider's cheap talk message can induce bidder separation because it rewards higher tender offers with higher takeover probabilities. A higher tender offer is costly to the bidder but will, in equilibrium, signal a higher type, inducing the insider to recommend a takeover more often because she wants to maximize the value of her shares. Of course, this presumes that outside shareholders indeed follow the insider's recommendation. Though the interests of the insider and the outside shareholders will never be perfectly aligned in equilibrium, outside shareholders follow the insider's recommendation in the fully separating equilibrium.

Absent communication between the insider and the outside shareholders, all bidder types conducting a takeover will pool on the same tender offer. Because bidder separation is a prerequisite for efficiency, our results imply that communication between shareholders is necessary for efficiency.

If ownership and information are separated, the efficient allocation of control rights is attainable in equilibrium because the bidder and insider reap all the gains from the takeover on all but the shares that the outside shareholders retain. In equilibrium, in a successful takeover, the bidder offers the outside shareholders exactly their outside option of the target's expected stand-alone value conditional on the insider's recommendation. Hence, the bidder can extract all gains from the takeover on the shares he acquires. Because the insider does not tender any shares, she cares not about the tender offer per se, but only about the information it contains. If the bidder fully reveals his private information via the tender offer, the insider has sufficient information to make efficient recommendations. She also wants to provide efficient recommendations since, if followed by the outside shareholders, these

[1] We use the terms share value and security benefits interchangeably in the paper.

maximize the value of the insider's share endowment. Because the informed entities obtain the full surplus created by the takeover on their respective shares, they face the correct incentives to maximize aggregate welfare.

By contrast, if the insider were the sole owner, she would need to sell part of her shares for a transfer of control and, thus, she would directly care about the tender offer itself. The tender offer will always be strictly below the post-takeover security benefits in a fully separating equilibrium because bidders would make zero profits otherwise, giving higher bidder types an incentive to imitate lower types. As a result, the insider does not obtain the full security benefits on the tendered shares in a takeover. Thus, if post-takeover security benefits do not exceed the target's stand-alone value by a large enough margin, the tender offer will be too low such that the insider prefers to remain in control, preventing some value-increasing takeover from being realized.

In equilibrium, the insider's recommendation only informs outside shareholders about whether the target's stand-alone value is above or below the post-takeover security benefits. By keeping shareholders (partially) in the dark, the insider can pool cases in her recommendation where outside shareholders would prefer to tender and not to tender under full information. Thus, the fact that outside shareholders only receive coarse information from the insider is necessary for efficiency.

In light of this result, our model calls into question mandatory disclosure requirements for provision of additional information when shareholders are approached with a tender offer. These provisions, which require disclosures either by management itself or by an external expert, are prevalent, and management has a strong incentive to comply to limit litigation risk.[2] Since mandatory disclosure requirements can undermine the coarseness of the information provided by the insider, they can undermine the separation of ownership and information and, thus, harm the efficient allocation of control rights.

One interpretation of the model is that the insider is the incumbent management. In this case, private information by the insider is pervasive, and incumbent management is compelled by law to give a "tender/keep" recommendation during takeovers.[3] If the current management is retained[4] in some capacity by the company after a takeover, she may even be committed *not* to tender any shares. Because the insider's communication strategy is endogenous to her tendering decision and because tendering any shares biases management against the takeover, not all value-increasing takeovers can be realized if management sells shares during the takeover. Hence, retaining management can prevent inefficient equilibria with "insider tendering."

[2] See Grossman and Hart (1980a); Bainbridge (1999); Becht, Bolton, and Röell (2003); and Kisgen, Qian, and Song (2009) for a discussion of mandatory disclosure and fairness opinions during takeovers. We provide a more detailed discussion at the end of Section IIIA.

[3] In the United States, incumbent management needs to respond to a tender offer with a recommendation in a Schedule 14D-9 Securities and Exchange Commission (SEC) filing.

[4] Management is frequently retained during takeovers. Estimates of management retention vary from about 30 percent to about 70 percent depending on sample and deal specifics (see, e.g., Wulf and Singh 2011; Qiu, Trapkov, and Takoub 2014; and Bargeron et al. 2017). Note that incumbent management can always retain her shares even she is not kept by the company.

Our model highlights how incumbent managers influence takeover outcomes even though outside shareholders hold the majority of formal control rights.[5] Their inside information gives them, in equilibrium, effective control over the takeover outcome. Hence, even absent takeover defense mechanisms such as poison pills that grant managers direct influence, incumbent management can determine takeover outcomes through simple cheap talk recommendations. Consistent with our model, empirical evidence suggests that takeovers are substantially more likely to succeed if they are endorsed by management; see, e.g., Schwert (2000); Baker and Savaşoglu (2002); and Bange and Mazzeo (2004).

Related Literature.—The literature on the market for corporate control has identified collective action problems and informational asymmetries that undermine the efficient allocation of control rights (Grossman and Hart 1980b; Shleifer and Vishny 1986; Marquez and Yılmaz 2008).[6] As Bagnoli and Lipman (1988), we focus on a company owned by a finite number of shareholders such that the free-rider problem identified by Grossman and Hart (1980b) does not impede the control allocation.

We show that cheap talk communication among shareholders can be an effective screening device of external bidders. In this way, our model is related to Hirshleifer and Titman (1990) and Burkart and Lee (2015), both of whom focus on private information of the external bidder. In Hirshleifer and Titman (1990), there exist mixed equilibria in which the bidder fully reveals his private information. In our setting, mixed tendering strategies by outsider shareholders cannot induce bidder separation. Burkart and Lee (2015) show how bidders can separate themselves if they can commit to relinquishing (part of) their private benefits of control. Levit (2017) studies the relation of cheap talk communication and tender offers. In contrast to our model, the bidder in Levit (2017) does not possess private information, which shuts down signaling—the main driver of our results.

Marquez and Yılmaz (2008) analyze a framework in which current shareholders have private information vis-à-vis the bidder, leading to a lemons problem. Ekmekci and Kos (2016) are able to resolve this issue by introducing a large minority shareholder whereas Ekmekci and Kos (2014) allow for information acquisition by the bidder and the shareholders. Marquez and Yılmaz (2012) compare public signals with information dispersed across shareholders and study the effects on the tender offer. Ekmekci, Kos, and Vohra (2016) derive the optimal mechanism for the sale of a company in a situation in which the buyer privately knows both the security benefits he will create and his private benefits of control. Also in a mechanism design setting, Ferreira, Ornelas, and Turner (2015) consider a situation in which two potential managers have private information about their ability to run a company they jointly own. While the mechanism in Ferreira, Ornelas, and Turner (2015) can allocate control and cash flow rights separately, in our model, control and cash flow rights need to be traded together, as is the case in practice for tender offers. Bernhardt, Liu, and Marquez (2018) introduce heterogeneous private valuations of investors in a takeover model and study the consequences for the tender offer characteristics.

[5] We distinguish here between formal control rights and effective control in the spirit of Aghion and Tirole (1997) and Burkart, Gromb, and Panunzi (1997).

[6] See Burkart and Panunzi (2008) for a detailed review of the literature.

Ordóñez-Calafí and Thanassoulis (2020) study the feedback loop between blockholders' selling decisions and takeover outcomes based on the premise that share sales reduce the target board's ability to resist the takeover.

In the context of mergers, Hansen (1987); Berkovitch and Narayanan (1990); and Eckbo, Giammarino, and Heinkel (1990) study a setting in which separation can be obtained by a mix of cash and equity offers. They consider two companies that want to exploit synergies of a merging assets, leading to a lemon's problem.

Up to now, a plethora of papers have analyzed strategic communication à la Crawford and Sobel (1982) in manifold economic environments. Antic and Persico (2017, 2020) provide a model of information transmission between an informed shareholder and an uninformed, controlling shareholder. The bias in the cheap talk stage depends on the share holdings, which are determined endogenously through trading in a competitive market prior to the communication stage. As a result, perfect information transmission is obtained in equilibrium. In our model, the conflict of interest is also endogenous. However, it is not only determined by the communicating shareholders but also through the tender offer of the external bidder. Full information transmission is not desirable in our model. Malenko and Tsoy (2019) show that advisors in English auctions who are biased toward overbidding can increase expected revenues and allocative efficiency via cheap talk messages. In their paper, cheap talk advice precedes the bidders' optimal price offer, whereas in our model, the bidder's price offer precedes the cheap talk message. Other papers studying strategic communication in a corporate governance context include Adams and Ferreira (2007); Almazan, Banerji, and Motta (2008); Harris and Raviv (2008); Kakhbod et al. (2019); Malenko (2014); Chakraborty and Yılmaz (2017); Levit (2019); Levit (2020); and Bittner et al. (2021).

I. Model

Environment.—An external bidder E considers the acquisition of a company. The target company has a continuum of shares of measure one outstanding. The bidder issues a tender offer $p_E \in \mathbb{R}_+$. For a successful takeover, he must acquire at least a fraction $\lambda > 0$ of the outstanding shares. The offer is unrestricted and conditional: if fewer than λ shares are tendered, the offer becomes void. If more than λ shares are tendered, E buys all shares tendered to him at per share price p_E.

The company has $j \in \{1, \ldots, J\}$ outside shareholders. A typical outside shareholder j owns a fraction s_j of the firm's shares. The outside shareholders jointly hold the majority of shares such that $\sum_{j=1}^{J} s_j = 1 - s > \lambda$.

A minority fraction $s \in (0, \lambda]$ of the shares are owned by an inside shareholder (I). One may think of the insider as the incumbent manager or as an actively involved long-term shareholder. As is often the case in practice, the current insider is endowed with only a minority of formal control rights but has effective control over the current management of the company. For ease of exposition, we start by assuming that I cannot tender any of her shares. Later, we show that not tendering any shares can indeed be a best response for I.

The game has three periods $t \in \{1,2,3\}$, and there is no discounting. At $t = 1$, E posts his tender offer p_E. At $t = 2$ and after observing p_E, I sends a cheap talk message m_I. At $t = 3$, given p_E and m_I, any outside shareholder j decides individually

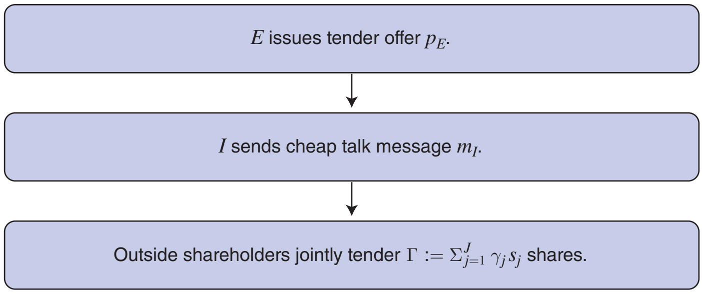

Figure 1. Timing

which fraction $\gamma_j \in [0,1]$ of her s_j shares to tender. Neither can I commit to tell the truth nor can the outside shareholders commit to a tender rule ex ante. The timing of events is summarized in Figure 1.

Information and Firm Value.—The firm value π has a common component $\omega \in (0,\infty)$ that accrues independently of the controlling entity. Because the common component is not specific to either controlling entity, we assume that it is common knowledge, for instance because it is likely to be priced in the current stock price. What makes the allocation problem potentially difficult is that whether a takeover is socially efficient depends on both the bidder's and the insider's private information. The bidder privately observes his type ω_E, which comprises information about his ability to run the company after a successful takeover. Furthermore, I has private information ω_I about the company's future profits if she remains in control.[7] The outside shareholders do not know either of the two types but, like all agents, observe ω. The bidder's and the insider's idiosyncratic types are independently distributed on $[0,1]$ according to continuous and commonly known cumulative distribution functions (CDFs) F_E and F_I. Both CDFs admit densities f_E and f_I with full support. The fact that the types are distributed according to different CDFs allows us to capture differences in expected firm values and uncertainty associated with I and E. The firm value π is then given by

$$\pi = \begin{cases} \omega + \omega_E, & \text{if takeover successful} \\ \omega + \omega_I, & \text{otherwise.} \end{cases} \tag{1}$$

While the private types are independent, the firm value is correlated under both controlling entities since $\omega > 0$. The degree of correlation depends on the size of ω.

Payoffs.—If no takeover occurs, outside shareholder j earns $\omega + \omega_I$ per share, irrespective of her tendering decision. Conversely, if the takeover succeeds, tendering γ_j

[7] Even though I already controls the company at the time of the tender offer, she will typically possess private, *inside* information. She may, for instance, know about an R&D project, or secret negotiations with a customer for which she is pivotal.

of her s_j shares yields p_E per sold share and security benefits of $\omega + \omega_E$ per retained share. An outside shareholder j's utility is thus

$$(2) \qquad v_j = \begin{cases} s_j\big(\gamma_j p_E + \big(1 - \gamma_j\big)[\omega + \omega_E]\big), & \text{if takeover successful} \\ s_j[\omega + \omega_I], & \text{otherwise.} \end{cases}$$

The insider's utility is given by the value of her shares under either control allocation

$$(3) \qquad u_I = \begin{cases} s[\omega + \omega_E], & \text{if takeover successful} \\ s[\omega + \omega_I], & \text{otherwise.} \end{cases}$$

The impact of private benefits of control for I is analyzed in Section IIIC. Observe that the level of disagreement between the insider and outside shareholders and, therefore, the scope for communication are endogenous to the bidder's tender offer. While the outside shareholders' payoff depends on the tender offer p_E directly, I is exclusively concerned with the post-takeover security benefits $\omega + \omega_E$. The bidder's utility is given by

$$(4) \qquad u_E = \begin{cases} \Gamma[\omega + \omega_E - p_E], & \text{if takeover successful} \\ 0, & \text{otherwise.} \end{cases}$$

The external bidder E derives constant utility normalized to zero if no takeover occurs. If the takeover is successful, E buys a fraction of $\Gamma = \sum_{j=1}^{J} \gamma_j s_j \geq \lambda$ shares from the outside shareholders at per-share costs of p_E and receives $\omega + \omega_E$ per share.

Model Applications.—In our model, outside shareholders will consider their impact on the takeover outcome. One can think of the company as a public company in which the majority of shares are owned by large shareholders. Due to their size, large shareholders such as institutional investors (hedge funds, mutual funds, etc.) or wealthy individuals have a nonnegligible impact on the takeover outcome in practice. Empirically, multiple large shareholders (blockholders) are prevalent in almost all companies, and they jointly hold large stakes. Edmans and Holderness (2017, p. 542) conclude in their survey that "[b]lockholders are ubiquitous. Virtually every corporation, of every size, in every country has them."

Observe that our model requires only that the majority of shares are owned by shareholders who consider their impact on the takeover outcome. One can add dispersed, atomistic shareholders to the model without affecting the results as long as large shareholders jointly hold the majority of shares. The reason is that small shareholders will free ride on the tendering decision of the large shareholders such that they do not affect the success of a takeover.

Another interpretation of the firm in our model is that of a more closely held private firm approached by an external bidder, such as a private equity fund. In this case, the private equity fund offers to buy out certain investors less involved in the management of the company, corresponding to the outside shareholders in our model. By contrast, in practice, there also needs to be an insider who manages the company on a daily basis, giving her access to private information vis-à-vis the less involved investors. Consistent with our model, empirical evidence suggests that private equity funds often retain incumbent managers,[8] committing them not to sell their ownership in the company.

Strategies.—Given the observed tender offer p_E and I's message m_I, a (pure) strategy for shareholder j specifies a fraction γ_j of tendered shares, i.e., $\gamma_j : \mathbb{R}_+ \times M_I \rightarrow [0,1]$ where $M_I = [0,1]$ is the message space. Denote the vector of outside shareholders' tendering decision by $\boldsymbol{\gamma} := (\gamma_1, \ldots, \gamma_J)$. Given p_E and ω_I, I chooses a message, i.e., $m_I : \mathbb{R}_+ \times [0,1] \rightarrow [0,1]$. Finally, a (pure) strategy for the bidder $p_E : [0,1] \rightarrow \mathbb{R}_+$ specifies a tender offer for any type ω_E. Throughout this paper, we assume that indifference of the outside shareholders is broken in favor of a takeover.[9] Our solution concept is the perfect Bayesian equilibrium in pure strategies, henceforth referred to as equilibrium. An equilibrium requires that (equilibrium objects are denoted with a star):

(i) given p_E^* and m_I^* and the tendering strategies of the other shareholders $\boldsymbol{\gamma}_{-j}^*$, any outside shareholder j chooses γ_j^* to maximize $\mathbb{E}\left[v_j \,|\, p_E^*, m_I^*, \boldsymbol{\gamma}_{-j}^*\right]$;

(ii) given all other strategies and ω_I, I chooses m_I^* to maximize her expected utility;

(iii) given all other strategies and ω_E, E chooses p_E^* to maximize his expected utility;

(iv) whenever possible, all players update their posterior beliefs according to Bayes' rule.

First-Best Allocation.—In our setting, ex post efficiency requires that the entity (E or I) with the higher type controls the company.

DEFINITION 1: *We call any equilibrium efficient or first best if a takeover occurs if and only if* $\omega_E \geq \omega_I$.

II. Unity of Ownership and Control

Suppose that ownership and control are not separated, i.e., the company is entirely owned by a single inside shareholder I such that $s = 1$. By exercising control, I privately knows ω_I. That is, the unity of ownership and control leads to a unity of ownership and information. As a result, the informed insider needs to tender $\lambda > 0$ of her shares for a takeover, which will make the optimal control allocation infeasible.

[8] Bargeron et al. (2017) show that incumbent management is retained in some capacity in roughly 70 percent of private equity acquisitions.

[9] This assumption is made to circumvent an openness problem and to ensure existence of equilibria.

Let $\eta \in [0,1]$ denote the fraction of her shares *I* tenders. For a given tender offer p_E and associated expected security benefits $\omega + \mathbb{E}[\omega_E | p_E]$, *I* wants to induce a takeover whenever there is some $\eta \geq \lambda$ such that

$$\eta p_E + (1-\eta)\big(\omega + \mathbb{E}[\omega_E | p_E]\big) \geq \omega + \omega_I. \tag{5}$$

Given the equilibrium tendering decision of *I* and his private type ω_E, *E* chooses a price $p_E \in \mathbb{R}_+$ to maximize his expected utility. The following proposition establishes that, in any equilibrium, the bidder's tender offer and the inside shareholder's tendering decision are jointly inconsistent with the first-best allocation of control rights.

PROPOSITION 1: *If ownership and control are not separated, there is no equilibrium with the first-best allocation of control rights.*

To obtain the first-best allocation, *I*'s tendering decision (5) must induce a takeover if and only if $\omega_E \geq \omega_I$. To this end, $p_E(\omega_E)$ must fully reveal ω_E because otherwise the insider does not have sufficient information to make the efficient tendering decision. Moreover, it has to hold that $p_E(\omega_E) = \omega + \omega_E$ such that *E* makes zero profits. Too see why, consider a fully separating equilibrium where this is not true. Then, the tender offer has to be strictly below the security benefits because $p_E(\omega_E) > \omega + \omega_E$ would induce strictly negative bidder profits. Because $p_E(\omega_E) < \omega + \omega_E$, the insider needs to give up part of the security benefits to the bidder on the shares she tenders for a successful takeover. By contrast, *I* enjoys the full security benefits if she remains in control. This asymmetry endogenously biases the insider against the takeover. As a result, if ω_I and ω_E are "close," the insider forgoes takeover opportunities that would improve firm value because the tender offer is too low.[10] Hence, an efficient equilibrium under a unity of ownership and control would require the tender offer to match the post-takeover security benefits.

However, zero bidder profits induced by $p_E(\omega_E) = \omega + \omega_E$ cannot be part of an efficient equilibrium. The reason is that higher types would have an incentive to imitate price offers of lower types who, in any efficient equilibrium, also have to have a positive probability of taking over. Hence, this imitation would secure strictly positive profits for any bidder type $\omega_E > 0$. We can therefore conclude that if ownership and control are not separated, efficient trade in the market for corporate control is not feasible.

REMARK 1: *If all shares must be traded for a change in control* (*i.e.*, $\lambda = 1$), *Proposition 1 follows from the classical impossibility result in bilateral trade by Myerson and Satterthwaite* (*1983*). *For* $\lambda < 1$, *Proposition 1 does not follow from Myerson and Satterthwaite* (*1983*) *because interdependent values arise in our setting. The insider participates in the expected value improvement by the bidder on*

[10]This point can further be illustrated with an example. Suppose $\omega_I \sim \mathcal{U}[0,1]$ and $\lambda = 1$. Then, one can show that there exists an equilibrium with $p_E^*(\omega_E) = \omega + \frac{1}{2}\omega_E$. A takeover occurs if and only if $\omega_E \geq 2\omega_I$—that is, the bidder's type must be twice as high as the insider's type to enable a successful takeover. This example shows that while bidder separation can be feasible in equilibrium, the resulting control allocation may still be far from efficient.

the shares she retains. Hence, there is some degree of alignment of interests among the inside shareholder and the external bidder that may give rise to efficient trade. Proposition 1 shows, however, that ex post efficiency is still not attainable in our setting.

III. Separating Ownership from Control and Information

The insider obtains private information by exercising control. Hence, if ownership and control are separated, ownership and information are separated as well. This separation implies that the insider can limit herself to advising outside shareholders on their tendering decisions, which will enable efficient trade in the market for corporate control. We also demonstrate how communication between shareholders is necessary for bidder separation.

Coordination Failure.—Since the company is owned by multiple outside shareholders, it may be the case that no single outside shareholder can individually enable a takeover—that is, $s_j < \lambda$ for all j. As a result, there exist equilibria exhibiting the following coordination failure: if an outside shareholder expects all other outside shareholders not to tender, her decision does not influence the outcome of the takeover, and thus, she may as well not tender. In equilibrium, no outside shareholder ever tenders.[11] To construct the efficient equilibrium we will focus on pure strategy equilibria in which outside shareholders are pivotal in their tendering decision, as in Bagnoli and Lipman (1988): if any outside shareholder tenders fewer than the equilibrium amount of shares, the takeover fails.

A. *Communication, Bidder Separation, and Efficiency*

In $t = 3$, the outside shareholders decide individually, given p_E, m_I, and the associated posteriors, how many shares to tender. In the efficient equilibrium, any outside shareholder j will be pivotal with all the shares she tenders to avoid free-riding incentives. Suppose, given the other outside shareholders' tender decisions, outside shareholder j needs to tender $\gamma_j > 0$ of her shares to induce a successful takeover. Then, j prefers tendering γ_j shares to letting the takeover fail if

$$\gamma_j p_E + \left(1 - \gamma_j\right)\left(\omega + \mathbb{E}[\omega_E | p_E]\right) \geq \omega + \mathbb{E}[\omega_I | m_I]. \tag{6}$$

In $t = 2$, given ω_I and p_E, I is indifferent between a takeover and no takeover if $\omega + \mathbb{E}[\omega_E | p_E] = \omega + \omega_I$. Thus, I endorses a takeover whenever

$$\omega_I \leq \omega_I^*(p_E) := \mathbb{E}[\omega_E | p_E]. \tag{7}$$

The indifference type ω_I^* equals the posterior expected type of E and is thus a function of p_E. When it is clear from the context, we drop the price. For any p_E and induced posterior belief about ω_E, the common support assumption ensures that

[11] This relies on the conditional form of the offer, which becomes void if a total fraction less than λ shares is tendered.

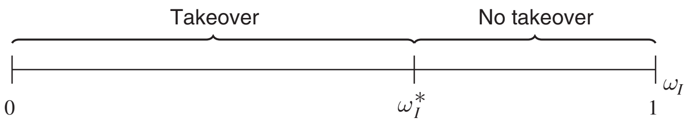

Figure 2. Cutoff Type ω_I^*

there is a unique cutoff type $\omega_I^* \in [0,1]$ at which the insider is indifferent. Note that I does not care about the tender offer directly but only about the information it contains about the post-takeover share value.

Figure 2 illustrates I's (informative) cheap talk messages. If the insider observes a low ω_I, and if she has a sufficiently high posterior expectation about the bidder's type, she prefers the outside shareholders to tender. Conversely, if I knows ω_I to be high relative to $\mathbb{E}[\omega_E | p_E]$, she advises the outside shareholders not to tender.

Bidder's Payoff.—In $t = 1$, if the outside shareholders follow I's recommendation, E's expected utility is given by

$$F_I\big(\omega_I^*(p_E)\big)\Gamma(p_E, m_I)[\omega + \omega_E - p_E]. \tag{8}$$

When the bidder chooses the tender offer p_E in $t = 1$, I's message m_I is not known since it is a function of her private information ω_I. External bidder E's expected utility thus equals the product of the probability that I's type is below the cutoff type $F_I\big(\omega_I^*(p_E)\big)$, the amount of shares tendered $\Gamma(p_E, m_I)$, and the profit earned on each acquired share $\omega + \omega_E - p_E$. Theorem 1 shows how the insider's cheap talk message can induce bidder separation. Theorem 1 also characterizes the associated equilibrium which implements the first-best control allocation.

THEOREM 1: *There is an equilibrium in which E fully reveals his type by posting*

$$p_E^*(\omega_E) = \omega + \mathbb{E}\big[\omega_I | \omega_I \leq \omega_I^*(\omega_E)\big].$$

Furthermore,

(*i*) *if* $\omega_I \leq \omega_I^*(\omega_E)$, *then* $m_I^* \in \big[0, \omega_I^*(\omega_E)\big]$, *and a takeover occurs with probability one*;

(*ii*) *if* $\omega_I > \omega_I^*(\omega_E)$, *then* $m_I^* \in \big(\omega_I^*(\omega_E), 1\big]$, *and a takeover occurs with probability zero*;

(*iii*) $\Gamma^*\Big(p_E^*, m_I^*\big(\omega_I \leq \omega_I^*(\omega_E)\big)\Big) = \lambda$;

(*iv*) *the equilibrium induces the first-best allocation of control rights.*

Theorem 1 establishes that there exists an equilibrium in which the bidder fully reveals his type via his tender offer. On the equilibrium path, given p_E^*, the insider's posterior belief assigns probability one to the true bidder type. Hence, I's indifference type becomes $\omega_I^* = \omega_E$. The insider sends a binary cheap talk message in favor of or against the takeover depending on whether $\omega_I \leq \omega_E$ or $\omega_I > \omega_E$. The outside shareholders find it optimal to follow I's message given p_E^* and their posterior beliefs of ω_E and ω_I. If a takeover occurs, outside shareholders jointly tender as few shares as possible, i.e., $\Gamma^*\big(p_E^*, m_I^*(\omega_I \leq \omega_I^*(\omega_E))\big) = \lambda$. In the following, we convey the intuition underlying the equilibrium in three steps and conclude by analyzing the case in which I can also tender her shares.

Tender Offer.—Inside shareholder I's cheap talk message enables bidder separation by introducing a way to compensate higher bidder types for posting higher prices. To see this, consider E's per share profit $F_I\big(\omega_I^*(p_E)\big)[\omega + \omega_E - p_E]$. If $\omega_I \leq \omega_I^*(p_E)$, I recommends a takeover, and if outside shareholders follow I's message, the takeover probability is given by $F\big(\omega_I^*(p_E)\big)$. Since $\omega_I^*(p_E) = \mathbb{E}[\omega_E | p_E]$, the takeover probability strictly increases in the expected post-takeover security benefits induced by E because these make the takeover more attractive to I. A higher tender offer is rewarded by a higher takeover probability but is also costly such that only higher types are incentivized to post higher tender offers, enabling bidder separation.

Formally, for a fully separating equilibrium to exist, there has to be a strictly increasing (and thus invertible) function $p_E : [0,1] \to \mathbb{R}_+$ such that, given any ω_E, when E chooses his bid[12] $p \in \mathbb{R}_+$ optimally, we have

$$p = p_E(\omega_E) \in \arg\max F_I\Big[\omega_I^*\big(p_E^{-1}(p)\big)\Big](\omega + \omega_E - p). \tag{9}$$

For any ω_E, this maximization yields the bidder-optimal bid given that the outside shareholders and the insider form posterior expectations according to $p_E(\omega_E)$, and the outside shareholders follow I's message. For any particular bid p, the takeover probability is thus determined by $F_I\Big[\omega_I^*\big(p_E^{-1}(p)\big)\Big]$. The unique solution to (9) is $p_E^*(\omega_E) = \omega + \mathbb{E}\big[\omega_I | \omega_I \leq \omega_I^*(\omega_E)\big]$, where, in the fully separating equilibrium, $\omega_I^* = \omega_E$. It is then easy to verify that, given that the insider and the outside shareholders form beliefs according to $p_E^*(\omega_E)$, it is indeed optimal for any type $\omega_E \in [0,1]$ to bid $p = p_E^*(\omega_E)$ relative to any other bid $p \in \big[p_E^*(0), p_E^*(1)\big]$.

Moreover, no bidder type wants to deviate to an (off-path) tender offer above $p_E^*(1)$ if off-path beliefs exclude all types who would strictly lose relative to their equilibrium payoff, i.e., if off-path beliefs satisfy the intuitive criterion by Cho and Kreps (1987). A deviation to some $p_E^{dev} > p_E^*(1)$ can only be profitable if $\Gamma\big(p_E^{dev}\big) > \lambda$ because $p_E^*(1)$ already ensures a takeover with probability one. $\Gamma\big(p_E^{dev}\big) > \lambda$ requires that $\omega + \mathbb{E}\big[\omega_E | p_E^{dev}\big] \leq p_E^{dev}$ such that off-path beliefs need to assign positive probability to types making at most zero profits. However, on the equilibrium path, all $\omega_E > 0$ make strictly positive expected profits. Further, as $p_E^*(0) = \omega$, the equilibrium can be supported by off-path beliefs, which assign probability one to $\omega_E = 0$ if E posts a price below the minimal firm value of ω.

[12] We introduce the notation of p here to distinguish between the bid function p_E and a specific bid p (number).

Under such off-path beliefs, any deviation to a tender offer below ω fails and, therefore, results in zero profits.

Observe that (9) considers only the per share profits. It is sufficient to solve for the bidder's per share profit because, in the proposed equilibrium, the total amount of tendered shares equals λ—independent of the posted price $p \in \left[p_E^*(0), p_E^*(1)\right]$. Jointly tendering the minimal amount of shares is the unique best response for outside shareholders because, for any $\omega_E \in (0,1]$, $p_E^*(\omega_E) = \omega + \mathbb{E}[\omega_I | \omega_I \leq \omega_E] < \omega + \omega_E$. In particular, because the post-takeover security benefits $\omega + \omega_E$ exceed the tender offer $p_E^*(\omega_E)$ for all $\omega_E > 0$, the outside shareholders would like to grant E control without selling their shares. As this is not feasible, they jointly sell the minimal amount of λ shares required for a transfer of control. If Γ were larger than λ, any shareholder tendering a positive amount of shares could increase profits by selling fewer shares while still enabling the takeover. The inequality $p_E^*(\omega_E) \leq \omega + \omega_E$ also guarantees that all bidder types at least receive their outside option of zero profits.

Cheap Talk Constraints.—In the equilibrium constructed in Theorem 1, the outside shareholders follow I's recommendation. If I recommends a takeover at some $p_E^*(\omega_E)$, i.e., $m_I^*(\omega_I \leq \omega_I^*)$, it is indeed optimal for any outside shareholder to tender $\gamma_j > 0$ shares to make the takeover successful if

$$(10) \quad \gamma_j p_E^*(\omega_E) + (1 - \gamma_j)\left(\omega + \mathbb{E}\left[\omega_E | p_E^*(\omega_E)\right]\right) \geq \omega + \mathbb{E}\left[\omega_I | \omega_I \leq \omega_I^*(\omega_E)\right].$$

Observe that the equilibrium tender offer, $p_E^*(\omega_E) = \omega + \mathbb{E}[\omega_I | \omega_I \leq \omega_E]$, is precisely the outside shareholders' outside option of leaving I in charge given that the insider sends a message in favor of a takeover. Hence, it is a best response for any outside shareholder j to tender γ_j shares if she is pivotal with all these shares and I recommends a takeover.

Conversely, suppose that I does not recommend a takeover at $p_E^*(\omega_E)$, i.e., $m_I^*(\omega_I > \omega_I^*)$. Then, $\gamma_j = 0$ is a best response for any outside shareholder since

$$(11) \quad \gamma_j p_E^*(\omega_E) + (1 - \gamma_j)\left(\omega + \mathbb{E}\left[\omega_E | p_E^*(\omega_E)\right]\right) < \omega + \mathbb{E}\left[\omega_I | \omega_I > \omega_I^*(\omega_E)\right],$$

which is slack in the proposed equilibrium. The reason is that in equilibrium, outside shareholders learn from I's opposition to the takeover that $\omega_I > \omega_I^* = \omega_E$ such that the right-hand side of (11) is larger than $\omega + \omega_E$. By contrast, since $p_E^*(\omega_E) < \omega + \omega_E$, the left-hand side of (11) is strictly smaller than $\omega + \omega_E$ for any $\gamma_j > 0$.

Note that none of the arguments relies on the coordination failure since our results also hold for $J = 1$.

Efficient Control Allocation.—Theorem 1 shows that the fully revealing equilibrium induces the first-best allocation of control rights and consequently is more efficient than any equilibrium under the unity of ownership and control (Section II). Given that E fully reveals his type in the equilibrium of Theorem 1, the intuition is straightforward: because $\omega_I^* = \omega_E$, I recommends a takeover if and only if it is efficient. Because the outside shareholders find it optimal to follow I's recommendation, the first-best allocation is obtained.

Since the outside shareholders receive exactly their outside option on the shares tendered, E obtains all expected gains he creates by taking control over the company on the shares he acquires. The insider, by keeping her shares, also obtains the entire surplus created by the trade on the shares she owns. Giving both privately informed entities the full gains of trade on their respective shares provides them with the correct incentives to implement the first-best control allocation. By contrast, the outside shareholders only participate in the bidder's value improvement via the $1 - s - \lambda$ shares that are not tendered.

The Role of Information.—Observe that the outside shareholders also care about the tender offer directly, whereas the insider only cares about the post-takeover security benefits. Since the equilibrium tender offer is smaller than the post-takeover security benefits, misaligned incentives prevent perfect information transmission in the efficient equilibrium: I's message merely reveals whether or not $\omega_I \leq \omega_I^* = \omega_E$.

Leaving the outside shareholders partly in the dark is key for efficiency. The reason is that I's cheap talk message pools two types of cases: those in which outside shareholders prefer to tender, and those in which outside shareholders would be better off not tendering if they had full information. In particular, when the post-takeover security benefits exceed the stand-alone value but the stand-alone value exceeds the tender offer, a takeover increases overall firm value and the insider's payoff. However, in such circumstances, a takeover reduces outside shareholders' payoff on each tendered share. To nonetheless induce the outside shareholders to tender, the insider's message needs to pool these cases with those cases in which the stand-alone value is below the tender offer such that outside shareholders are on average better off following the recommendation. Thus, first best is only attainable with partially informed outside shareholders.

Nevertheless, our model does not argue for more insider information in general. If I had no private information at the outset (i.e., ω_I was public information to all agents), an efficient equilibrium can be constructed. In this case, E could simply set the tender offer equal to the current shareholders' commonly known outside option of $\omega + \omega_I$ whenever $\omega_E \geq \omega_I$, inducing the first-best allocation. Rather, our model builds on the idea that insiders naturally obtain private information by exercising control. We then ask how to optimally deal with this inside information in the market for corporate control. A common regulatory response to the prevalence of inside information during takeovers has been the introduction of mandatory disclosure requirements[13] and the provision of so called fairness opinions.[14] These give outside shareholders the right to force the company's insiders to provide (additional) information after a tender offer. According to our theory, these regulations can harm the efficient allocation of control rights. In particular, our results suggest that *after* a tender offer has been issued, outside shareholders should not obtain more information than provided by the insider's cheap talk recommendation. An inefficiency is likely to arise if outside shareholders can enforce additional disclosure with the

[13] For a detailed discussion of mandatory disclosure requirements during takeovers, see, e.g., Grossman and Hart (1980b); Bainbridge (1999); and Becht, Bolton, and Röell (2003).

[14] Fairness opinions are external expert evaluations of firm value conducted by investment banks or consultancies after a tender offer. Kisgen, Qian, and Song (2009) show that fairness opinions are prevalent and present legal cases that imply that shareholders can force management to conduct fairness opinions.

threat of a lawsuit: at the time of their tendering decision in $t = 3$, outside shareholders could increase their payoff if they received more information than revealed by m_I^*. Eliminating outside shareholders' right to enforce additional disclosure may thus increase allocative efficiency.

Fairness opinions about the current value of the company have been described as uninformative rubber-stamping of management's recommendation (Kisgen, Qian, and Song 2009). Our model provides a rationale for these uninformative fairness opinions: provided that management is legally compelled to hire an expert to disclose additional information but has discretion over how informative said fairness opinion is, it would be optimal for the incumbent management (and overall welfare) to request uninformative fairness opinions.

Overall, our findings argue against disclosure requirements after tender offers, but not against a general reduction of insider information.

Insider Tendering.—Some insiders are not committed to staying with the company, e.g., if incumbent management is not retained after a takeover. Thus, the insider may optimally tender her shares to maximize her profits. To capture this idea, we add the possibility that I tenders any fraction $\eta \in [0,1]$ of her shares simultaneously with the outside shareholders. Inside shareholder I's tendering decision depends on her type ω_I and the tender offer p_E. Apart from tendering, the game evolves as in the baseline model, and we consider perfect Bayesian equilibria in pure strategies.

Inside shareholder I's tendering decision can potentially affect the outcome via two channels. First, it can change the total amount of shares tendered which is now given by $\Gamma = \eta s + \sum_{j=1}^{J} \gamma_j s_j$. Second, the anticipation of her trade can alter I's equilibrium recommendation.

PROPOSITION 2:

(*i*) *The equilibrium constructed in Theorem 1, together with I never tendering any shares, is an equilibrium in the extended game.*

(*ii*) *In any equilibrium in which for some* p_E, $\eta^*(p_E)s + \sum_{j=1}^{J} \gamma_j^*(p_E)\, s_j \geq \lambda$ *with* $\eta^*(p_E) > 0$, *not all value-increasing takeovers are realized.*

(*iii*) *The equilibrium constructed in Theorem 1, together with I never tendering any shares, maximizes expected insider profits.*

The proof is provided in the online Appendix. Proposition 2 establishes that even when trading is feasible, the combination of I never selling any shares and the strategies from Theorem 1 forms an equilibrium.

Further, Proposition 2 shows that not all value-increasing takeovers can be realized if I tenders shares. If she sells any shares, I cannot enjoy the full security benefits on these shares when E obtains control, effectively biasing her against the takeover. The reason is again that, in any equilibrium, the tender offer p_E is strictly below the expected security benefits $\omega + \mathbb{E}[\omega_E | p_E]$. In anticipation of her trade, I will thus adapt her equilibrium message as to recommend fewer takeovers.

The efficient equilibrium maximizes I's payoff because she obtains all the takeover gains on her share endowment, and all value-increasing takeovers are realized. Hence, insiders have an incentive to stay invested in the target even when relinquishing control.

B. *The Importance of Communication*

The efficient equilibrium constructed in Theorem 1 builds on cheap talk recommendations by the insider. This is consistent with the fact that incumbent management is obliged to issue a "tender/keep" recommendation in a Schedule 14D-9 filing for the US Securities and Exchange Commission (SEC) if approached with a tender offer. Is this communication between the insider and outside shareholders necessary for efficient trade in the market for corporate control?

To shed light on this question, we analyze the case in which I cannot communicate (or I's message is babbling[15]) in this section. To isolate the effect of communication, we again focus the analysis on the case in which I cannot tender any shares.

PROPOSITION 3: *In any babbling equilibrium,*

(*i*) *either a takeover never occurs*;

(*ii*) *or there exists a cutoff price* $\hat{p}_E < \omega + 1$ *such that:*
if $\omega_E < \hat{p}_E - \omega$, *a takeover occurs with probability zero*,
if $\omega_E > \hat{p}_E - \omega$, *E posts* $\hat{p}_E$, *a takeover occurs with probability one*, *and* $\Gamma^*(\hat{p}_E) = \lambda$;

(*iii*) *or* $\hat{p}_E = \omega + 1$, *a takeover occurs if and only if* $\omega_E = 1$, *and* $\Gamma^*(\hat{p}_E) \geq \lambda$.

The proof is provided in the online Appendix. Proposition 3 shows that in any equilibrium without informative communication, all bidder types that realize a takeover pool on the same tender offer. In particular, there exist three kinds of equilibria. The first type of equilibrium is the consequence of a coordination failure and exists if no shareholder individually holds a majority stake. If no outside shareholder ever tenders any shares, no shareholder is pivotal on her own. Hence, never selling any shares constitutes a best response.

Second, there are cutoff equilibria in which only bidder types above some cutoff conduct a takeover by pooling on the same tender offer $\hat{p}_E < \omega + 1$, whereas all smaller bidder types never take control. In these cutoff equilibria, shareholders jointly tender $\Gamma^*(\hat{p}_E) = \lambda$ shares whenever a takeover occurs. In principle, shareholders would like to free ride on the tendering decisions of other shareholders because the tender offer is strictly below the post-takeover security benefits, i.e., $\hat{p}_E < \omega + \mathbb{E}[\omega_E | \omega_E \geq \hat{p}_E - \omega]$. To overcome free riding, every shareholder needs to be pivotal with all the shares she tenders. If any shareholder tendered more shares such

[15] Since I's recommendation is cheap talk, there always exists an equilibrium in which her message is uninformative. A message $m_I(p_E, \omega_I)$ is uninformative (or *babbling*) if, for all $p_E \in \mathbb{R}_+$, $m_I(p_E, \omega_I)$ is independent of ω_I.

that $\Gamma(\hat{p}_E) > \lambda$, she would have a profitable deviation to tender fewer shares while still making the takeover successful.

As the first kind of equilibrium, the third type only exists if no shareholder individually holds a majority stake. Then, in equilibrium, for all $p_E < \omega + 1$, no shareholder ever tenders enough shares to make another shareholder pivotal. Thus, at any $p_E < \omega + 1$, selling no shares is a best response for all outside shareholders. $p_E = \omega + 1$ is only posted by $\omega_E = 1$ because all other types would make strictly negative profits. Because the post-takeover security benefits equal the tender offer and exceed the expected stand-alone value, shareholders are indifferent between any γ_j that makes the takeover succeed such that $\Gamma^*(p_E = \omega + 1) \in [\lambda, 1 - s]$.

Proposition 3 shows that without informative cheap talk, all bidder types that realize a takeover pool on the same tender offer. Since first best requires bidder separation and a strictly positive takeover likelihood for each $\omega_E > 0$, efficient trade in the market for corporate control is not feasible without meaningful communication between inside and outside shareholders.

In principle, bidder separation demands that higher types are incentivized to post higher tender offers. This could be achieved by granting E more shares upon a higher tender offer. Posting higher tender offers can only be profitable for higher bidder types if the tender offer is still below the expected post-takeover security benefits. However, when the tender offer is below the expected post-takeover security benefits, outside shareholders jointly tender as few shares as possible. Hence, bidder separation cannot be achieved by varying the amount of shares tendered in a successful takeover.

Alternatively, higher tender offers can be incentivized by increasing the ex ante takeover likelihood from E's perspective. In the efficient equilibrium of Theorem 1 this occurs because higher tender offers signal higher bidder types, inducing I to recommend a takeover more often. How can one vary the takeover likelihood without I's recommendations? Hirshleifer and Titman (1990) show that in a model with one-sided asymmetric information and completely dispersed ownership, full separation of the bidder may be attainable if target shareholders play mixed strategies.[16] However, even if we allowed for mixed strategies of the outside shareholders, separation among bidder types realizing a takeover is not feasible in our model. To see why, note that outside shareholders' outside option of leaving I in charge is fixed at $\omega + \mathbb{E}[\omega_I]$ without I's recommendations. By contrast, if higher bidder types post higher tender offers to separate, they strictly increase outside shareholders' profits in the takeover relative to lower tender offers. As a result, outside shareholders cannot be indifferent at different tender offers such that mixing cannot be a best response. See the online Appendix for a formal discussion.

C. *Private Benefits of Control*

Depending on the insider's identity, private benefits of control may arise. While active shareholders do not have a clear bias for or against a takeover, incumbent management may derive private benefits of control. Thus, in this section, we refer to

[16] It is noteworthy, however, that mixing will always cause a loss in welfare such that the first-best allocation is not feasible.

I as the incumbent management, and we extend the basic model to include private benefits of control for I such that

$$u_I = \begin{cases} s[\omega + \omega_E], & \text{if takeover successful} \\ s[\omega + \omega_I] + B_I, & \text{otherwise.} \end{cases} \tag{12}$$

Let $b_I := B_I/s$ denote I's private benefit per share such that I's indifference type between a takeover and no takeover is given by

$$\omega_I^{**} := \max\{\mathbb{E}[\omega_E | p_E] - b_I; 0\}. \tag{13}$$

Whenever $\omega_I \leq \omega_I^{**}$, the insider favors a takeover. We show in the online Appendix that, if b_I is small enough, there exists a fully separating equilibrium similar to Theorem 1. In this equilibrium, outside shareholders follow I's recommendation such that a takeover occurs if and only if $\omega_I \leq \omega_I^{**} = \max\{\omega_E - b_I; 0\}$. Hence, inefficiently few takeovers occur because the incumbent manager is biased against the takeover even if she does not tender any of her shares.

However, it is easy to see that efficiency can be restored if I is provided with an adequate golden parachute upon a change in control.[17] With a golden parachute of size $G \in \mathbb{R}_+$, I's utility becomes

$$u_I = \begin{cases} s[\omega + \omega_E] + G, & \text{if takeover successful} \\ s[\omega + \omega_I] + B_I, & \text{otherwise.} \end{cases} \tag{14}$$

As a result, whenever the golden parachute is set equal to the private benefits of control, i.e., $G = B_I$, I's indifference type is $\omega_I^{**}(p_E) = \max\{\mathbb{E}[\omega_E | p_E] - (B_I - G)/s; 0\} = \mathbb{E}[\omega_E | p_E]$. Hence, it equals the indifference type (7) from the baseline model such that there is an efficient equilibrium as in Theorem 1.

IV. Concluding Remarks

This paper identifies an upside of the separation of ownership and control. The key premise of our model is that by virtue of exercising control, insiders obtain private information. As a result, the separation of ownership and control leads to a separation of ownership and information. We show that efficient trade in the market for corporate control is only attainable if information and ownership are separated—in that the majority of shares are owned by outside shareholders. Our model highlights the importance of strategic communication between inside and outside shareholders. We characterize how this communication is shaped by tender offers and vice versa.

According to our model, it is key for efficiency that insiders only provide coarse information to outside shareholders. Because legally prescribed fairness opinions

[17] We are by no means the first to consider the problem of golden parachutes or severance pay. See, e.g., Lambert and Larcker (1985); Knoeber (1986); Harris (1990); Almazan and Suarez (2003); Eisfeldt and Rampini (2008); and Inderst and Müller (2010). However, none of these papers considers how golden parachutes influence management's advisory role in takeovers.

and mandatory disclosure requirements can undermine the coarseness of information, they can prevent efficient trade in the market for corporate control.

Consistent with the empirical evidence, our model shows how insiders' private information gives them effective control over takeover outcomes even if the majority of formal control rights is with outside shareholders.

Appendix

PROOF OF PROPOSITION 1:

Step 1: If E does not fully separate in an equilibrium, then first best is not achieved in this equilibrium.

Suppose, on the way to a contradiction, that this was not true, i.e., there exist bidder types ω_E, ω_E', with $\omega_E > \omega_E'$ but $p_E(\omega_E) = p_E(\omega_E')$ and first best is attained. By the common support assumption, there exists an open interval $(\underline{\omega}_I, \overline{\omega}_I) \neq \emptyset$ such that $(\underline{\omega}_I, \overline{\omega}_I) \subset (\omega_E', \omega_E)$. For all $\omega_I \in (\underline{\omega}_I, \overline{\omega}_I)$, first best requires that a takeover does not occur at ω_E', but does occur at ω_E. But since $p_E(\omega_E) = p_E(\omega_E')$, either a takeover occurs at both types or at none. Hence, first best cannot be achieved and we obtain a contradiction.

Step 2: If E fully separates, first best requires zero profits for all bidder types.

Suppose p_E perfectly reveals ω_E. Then, the shareholder prefers a takeover whenever there is some $\eta \geq \lambda > 0$ such that $\eta p_E(\omega_E) + (1-\eta)[\omega + \omega_E] \geq \omega + \omega_I$. Because $p_E(\omega_E) > \omega + \omega_E$ induces negative bidder profits, it cannot be part of an equilibrium. Further, for any $p_E(\omega_E) < \omega + \omega_E$, the shareholder's tendering decision implies that there is a $\omega_I \in \left(\lambda\left(p_E(\omega_E) - \omega\right) + (1-\lambda)\,\omega_E, \omega_E\right)$ such that no takeover occurs at $p_E(\omega_E)$ because $\eta p_E(\omega_E) + (1-\eta)[\omega + \omega_E] < \omega + \omega_I$ for any $\eta \geq \lambda$. However, first best requires a takeover since $\omega_I < \omega_E$. Hence, the optimal allocation rule is implemented in equilibrium if and only if $p_E(\omega_E) = \omega + \omega_E$ such that $u_E(\omega_E) = 0\ \forall \omega_E \in [0,1]$.

Step 3: There is no equilibrium in which first best is achieved.

Suppose first-best was attained in equilibrium such that, by steps 1 and 2, $p_E(\omega_E) = \omega + \omega_E$. But then any $\omega_E > 0$ could profitably deviate by imitating some type $\omega_E'' \in (0, \omega_E)$ by offering $p_E(\omega_E'') = \omega + \omega_E''$. After this deviation, the takeover probability $F_I(\omega_E'')$ is strictly positive because first best requires that a takeover occurs for all $\omega_I \in [0, \omega_E'')$, and by full support of f_I. Therefore, the proposed deviation gives strictly positive profits of $[\omega_E - \omega_E'']\,F_I(\omega_E'') > 0$ which yields a contradiction and completes the proof.

PROOF OF THEOREM 1:

Step 0: Single crossing and I's equilibrium message.

Given some tender offer p_E, $\mathbb{E}\left[u_I(\text{takeover})\,|\,p_E\right] = s\left(\omega + \mathbb{E}[\omega_E|p_E]\right)$ does not depend on ω_I whereas $\mathbb{E}\left[u_I(\text{no takeover})\,|\,p_E\right] = s(\omega + \omega_I)$ is strictly increasing in ω_I. Thus, all types $\omega_I \leq \omega_I^*$ prefer the takeover to succeed and all $\omega_I > \omega_I^*$ prefer the takeover to fail, where $\omega_I^* = \mathbb{E}[\omega_E|p_E]$.

Step 1: Necessary condition for a fully separating tender offer.

Suppose the bidder plays a fully separating strategy, i.e., p_E is strictly increasing (and thus invertible) function $p_E : [0,1] \to \mathbb{R}_+$. Let p denote a specific bid p (number) whereas p_E is the fully separating bid function. Further, suppose that $\Gamma^*\left(p_E, m_I^*\left(\omega_I \leq \omega_I^*(p_E)\right)\right) = \lambda$ which we verify below. Then, given his true type ω_E, the bidder's optimal bid p is given by

$$\underset{p \in \mathbb{R}_+}{\arg\max}\, F_I\left[\omega_I^*\left(p_E^{-1}(p)\right]\lambda[\omega + \omega_E - p]. \tag{A.1}$$

The first-order condition (FOC) is

$$f_I\left[\omega_I^*\left(p_E^{-1}(p)\right]\omega_I^{*\prime}\left(p_E^{-1}(p)\right)p_E^{-1}(p)'[\omega + \omega_E - p] - F_I\left[\omega_I^*\left(p_E^{-1}(p)\right)\right] = 0. \tag{A.2}$$

Observe that because p_E is strictly increasing, $\omega_I^* = \mathbb{E}[\omega_E|p_E] = \omega_E$. At the equilibrium bid $p = p_E(\omega_E)$, the FOC can be rewritten as the following ordinary differential equation (ODE):

$$p_E'(\omega_E) = \frac{f_I\left[\omega_I^*(\omega_E)\right]}{F_I\left[\omega_I^*(\omega_E)\right]}\left(\omega + \omega_E - p_E(\omega_E)\right) = \frac{f_I(\omega_E)}{F_I(\omega_E)}\left(\omega + \omega_E - p_E(\omega_E)\right). \tag{A.3}$$

The general solution to (A.3) is[18]

$$p_E(\omega_E) = \frac{\int_0^{\omega_E} f_I(z)(z + \omega)\,dz + C}{F_I(\omega_E)}, \tag{A.4}$$

where C is a constant that pins down the solution depending on the initial value. As the lowest bidder type is $\omega_E = 0$, $C = 0$ such that

$$p_E^*(\omega_E) = \frac{\int_0^{\omega_E} f_I(z)z\,dz}{F_I(\omega_E)} + \omega\frac{\int_0^{\omega_E} f_I(z)\,dz}{F_I(\omega_E)} = \mathbb{E}[\omega_I|\omega_I \leq \omega_E] + \omega. \tag{A.5}$$

[18] Applying Leibniz's integral rule and taking the derivative with respect to ω_E yields $p_E'(\omega_E) = \frac{f_I(\omega_E)\,\omega_E F_I(\omega_E) - \left(\int_0^{\omega_E} f_I(z)z dz + C\right)f_I(\omega_E)}{F^2(\omega_E)}$ which can be written as $\frac{f_I(\omega_E)}{F_I(\omega_E)}\left[\omega_E - \frac{\left(\int_0^{\omega_E} f_I(z)z dz + C\right)}{F(\omega_E)}\right]$. Plugging in (A.4) in (A.3), and comparing shows the claim.

Step 2: Sufficiency.

We now show that the bidder's objective function is concave evaluated at the equilibrium price function derived above and that any bidder type ω_E optimally chooses $p = p_E^*(\omega_E)$; i.e., $p_E^*(\omega_E)$ indeed constitutes an equilibrium price function. The objective of the bidder (up to the amount of shares he acquires that is independent of p_E), evaluated at $p_E^*(\omega_E)$, becomes

$$\text{(A.6)} \qquad F_I\left[p_E^{*-1}(p)\right]\left[\omega + \omega_E - \frac{\int_0^{p_E^{*-1}(p)}(\omega+\omega_I)f_I(\omega_I)\,d\omega_I}{F_I\left[p_E^{*-1}(p)\right]}\right]$$
$$= (\omega+\omega_E)\,F_I\left[p_E^{*-1}(p)\right] - \int_0^{p_E^{*-1}(p)}(\omega+\omega_I)f_I(\omega_I)\,d\omega_I.$$

To see that it is indeed optimal to post $p = p_E^*(\omega_E)$, denote $\hat{\omega}_E := p_E^{*-1}(p)$ such that the objective function becomes

$$\text{(A.7)} \qquad (\omega+\omega_E)\,F_I[\hat{\omega}_E] - \int_0^{\hat{\omega}_E}(\omega+\omega_I)f_I(\omega_I)\,d\omega_I.$$

Taking the derivative with respect to $\hat{\omega}_E$ yields

$$\text{(A.8)} \qquad f_I(\hat{\omega}_E)\left[(\omega+\omega_E) - (\omega+\hat{\omega}_E)\right] = f_I(\hat{\omega}_E)[\omega_E - \hat{\omega}_E],$$

which is zero at $\hat{\omega}_E = \omega_E$, strictly positive whenever $\hat{\omega}_E < \omega_E$, and strictly negative for $\hat{\omega}_E > \omega_E$. Hence, the bidder indeed finds it optimal to post $p = p_E^*(\omega_E)$ relative to any other $p \in \left[p_E^*(0), p_E^*(1)\right]$, given the other players expect him to play $p_E^*(\omega_E)$.

Since $p_E^*(\omega_E) = \omega + \mathbb{E}[\omega_I|\omega_I \leq \omega_E] < \omega + \omega_E$ implies strictly positive expected profits for $\omega_E > 0$ and zero for $\omega_E = 0$, E also prefers $p_E^*(\omega_E)$ relative to no takeover.

Step 3: Outside shareholders jointly tender never more than λ shares in a successful takeover.

Since $p_E^*(\omega_E) = \omega + \mathbb{E}[\omega_I|\omega_I \leq \omega_E] < \omega + \omega_E$ holds for all $\omega_E \in (0,1]$, if $\Gamma^* > \lambda$ were true, any shareholder j who tenders a nonzero fraction $\gamma_j > 0$ of her shares would have a profitable deviation to tender fewer shares while still making the takeover still succeed. Hence, in a successful takeover shareholders will jointly tender $\Gamma^* = \lambda$.

Step 4: Given p_E^* and $m_I^*(\omega_I \leq \omega_I^*(p_E^*))$, any $\boldsymbol{\gamma}$ such that $\Gamma^*(p_E^*, m_I^*(\omega_I \leq \omega_I^*))$ $= \lambda$ is a best response for outside shareholders.

By step 3, if a takeover occurs, any outside shareholder is pivotal with all her shares, i.e., $\Gamma^*(p_E^*, m_I^*(\omega_I \leq \omega_I^*)) = \lambda$. Then, any pivotal outside shareholder j indeed wants to make the takeover successful by tendering $\gamma_j^* > 0$ shares if

$$\text{(A.9)} \quad \gamma_j^* p_E^*(\omega_E) + (1-\gamma_j^*)\left(\omega + \mathbb{E}\left[\omega_E | p_E^*(\omega_E)\right]\right) \geq \omega + \mathbb{E}\left[\omega_I | \omega_I \leq \omega_I^*(\omega_E)\right].$$

Plugging in (p_E^*, ω_I^*) yields

$$\text{(A.10)}\quad \gamma_j^*(\omega + \mathbb{E}[\omega_I|\omega_I \leq \omega_E]) + (1-\gamma_j^*)(\omega + \omega_E) \geq \omega + \mathbb{E}[\omega_I|\omega_I \leq \omega_E],$$

which holds true for any $\gamma_j^* \in (0,1]$. Hence, if I recommends a takeover, it is a best response for any pivotal shareholder to tender sufficiently many shares to realize the takeover.

Step 5: Given p_E^* and $m_I^*(\omega_I > \omega_I^*(p_E^*))$, $\gamma = (0, \ldots, 0)$ is a best response for outside shareholders.

If p_E^* and $m_I^*(\omega_I > \omega_I^*)$ and no outside shareholder is pivotal on her own, it is a best response for any outside shareholder not to tender any shares. If p_E^* and $m_I^*(\omega_I > \omega_I^*)$, and an outside shareholder is pivotal, the pivotal outside shareholder does not want to tender $\gamma_j > 0$ shares to make the takeover successful if

$$\text{(A.11)}\quad \gamma_j p_E^*(\omega_E) + (1-\gamma_j)\left(\omega + \mathbb{E}\left[\omega_E|p_E^*(\omega_E)\right]\right) < \omega + \mathbb{E}\left[\omega_I|\omega_I \geq \omega_I^*(\omega_E)\right].$$

Plugging in p_E^* and ω_I^* yields

$$\text{(A.12)}\quad \gamma_j(\omega + \mathbb{E}[\omega_I|\omega_I \leq \omega_E]) + (1-\gamma_j)(\omega + \omega_E) < \omega + \mathbb{E}[\omega_I|\omega_I > \omega_E].$$

The right-hand side is strictly larger than the left-hand side by the full support assumption for any $\gamma_j > 0$.

Step 6: There are no profitable deviations to $p \notin \left[p_E^*(0), p_E^*(1)\right]$ given that off-path beliefs satisfy the intuitive criterion and assign probability 1 to $\omega_E = 0$ for any $p_E < \omega$.

Any deviation to $p_E^{dev} > p_E^*(1)$ cannot be profitable if off-path beliefs satisfy the intuitive criterion by Cho and Kreps (1987). As $F_I(p_E^*(1)) = 1$, a takeover occurs with certainty when the bidder posts the highest equilibrium price. Posting any price $p_E^{dev} > p_E^*(1)$ could therefore only be profitable if $\Gamma^*(p_E^{dev}) > \lambda$. This requires that off-path beliefs induce $p_E^{dev} \geq \omega + \mathbb{E}\left[\omega_E|p_E^{dev}\right]$ which, in turn, implies that off-path beliefs assign positive probability to types $\omega_E > 0$ that make weakly negative profits by the deviation. However, this is not consistent with the intuitive criterion since expected profits are strictly positive for all $\omega_E > 0$ on the equilibrium path.

Further, as $p_E^*(0) = \omega$, downward deviations to off-path prices are in $[0, \omega)$, i.e., below the minimal post-takeover value of the company. The proposed equilibrium can, for instance, be supported by off-path beliefs assigning $\omega_E = 0$ to any such deviation since then

$$\text{(A.13)}\qquad \gamma_j p_E^{dev} + (1-\gamma_j)\left(\omega + \mathbb{E}\left[\omega_E|p_E^{dev}\right]\right) < \omega \leq \omega + \mathbb{E}[\omega_I|m_I],$$

and no outside shareholder has an incentive to tender any shares (pivotal or not).

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